

5CM506 Data driven systems

Part 1

Data & Database

Learning Objectives

- Explain why humankind's interest in data goes back to ancient times.
- Describe how data needs have historically driven many information technology developments.
- Describe the evolution of data storage media during the last century.
- Relate the idea of data as a corporate resource that can be used to gain a competitive advantage to the development of the database management systems environment.
- Differentiate the types of databases and database applications
- Understand the principals of typical DBMS functionality
- Explain the main characteristics of the database approach
- Know the types of database users
- Appraise the advantages of using the database approach
- Summarize the historical development of database technology
- Know how to extend database capabilities
- Estimate when not to use databases

Outline



- Data in history
- Data storage media (today and in the past)
- Types of Databases and Database Applications
- Basic Definitions
- Typical DBMS Functionality
- Example of a Database (UNIVERSITY)
- Main Characteristics of the Database Approach
- Types of Database Users
- Advantages of Using the Database Approach
- Historical Development of Database Technology
- Extending Database Capabilities
- When Not to Use Databases

Data

- Data - the foundation of technological activity
- Database - a highly organized collection of assembled data
- Database Management System - sophisticated software that controls the database and the database environment

What is Data?

- A single piece of data is a single fact about something that interests us.
- A fact can be any characteristic of an object.

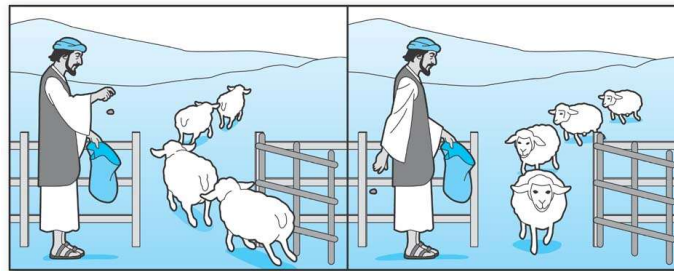
Salesperson Number	Salesperson Name	City	State	Office Number	Commission Percentage	Year of Hire
137	Baker	Detroit	MI	1284	10	1995

Facts about salesperson Baker.

History of Data

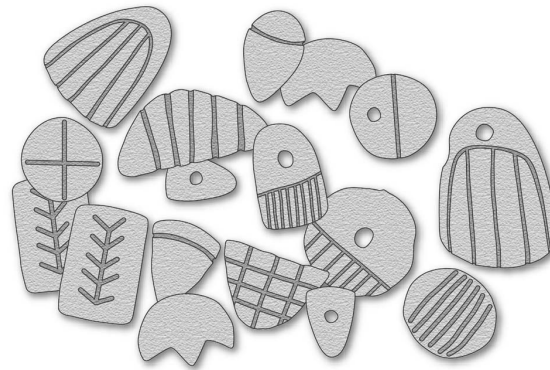
- People have been interested in data for at least the past 12,000 years.
- Non-computer, primitive methods of data storage and handling.

What is Data? (Cont.)



- Shepherds kept track of their flocks with pebbles.
- A primitive but legitimate example of data storage and retrieval.

History of Data



- Dating back to 8500 B.C., unearthed clay tokens or “counters” may have been used for record keeping in primitive forms of accounting.
- Tokens, with special markings on them, were sealed in hollow clay vessels that accompanied commercial goods in transit.

Data Through the Ages

- Record-keeping - the recording of data to keep track of how much a person has produced and what it can be bartered or sold for.
- With time, different kinds of data were kept
 - calendars, census data, surveys, land ownership records, marriage records, records of church contributions, family trees, etc.

History of Data



- Double-entry bookkeeping - originated in the trading centers of fourteenth century Italy.
- The earliest known example is from a merchant in Genoa and dates to the year 1340.

Early Data Problems Spawn Calculating Devices

- People interested in devices that could “automatically” process their data.
- Blaise Pascal produced an adding machine that was an early version of today’s mechanical automobile odometers.

Punched Cards - Data Storage

- Invented in 1805 by Joseph Marie Jacquard of France.
- Jacquard's method of storing fabric patterns, a form of graphic data, as holes in punched cards was a very clever means of data storage.
- Of great importance for computing devices to follow.

Era of Modern Information Processing

- The 1880 U.S. Census took about seven years to compile by hand.
- Basing his work on Jacquard's punched card concept, Herman Hollerith arranged to have the census data stored in punched cards and invented machinery to tabulate them.
- In 1896 Hollerith formed the Tabulating Machine Company to produce and commercially market his devices -- this later became IBM.

Era of Modern Information Processing

- James Powers developed devices to automatically feed cards into the equipment and to automatically print results.
- In 1911 he established the Powers Tabulating Machine Company -- this later became Unisys Corporation.

The Mid-1950s

- The introduction of electronic computers.
- Witnessed a boom in economic development.
- From this point onward, it would be virtually impossible to tie advances in computing devices to specific, landmark data storage and retrieval needs.

Modern Data Storage Media

- Punched paper tape - The earliest form of modern data storage, introduced in the 1870s and 1880s.
- Punched cards were the only data storage medium used in the increasingly sophisticated electromechanical accounting machines of the 1920s, 1930s, and 1940s.

Modern Data Storage Media

- Middle to late 1930s saw the beginning of the era of erasable magnetic storage media.
- By late 1940s, early work was done on the use of magnetic tape for recording data.
- By 1950, several companies were developing the magnetic tape concept for commercial use.

Modern Data Storage Media

- Magnetic Tape - commercially available units in 1952.
- Direct Access Magnetic Devices - began to be developed at MIT in the late 1930s and early 1940s.
- Magnetic Drum - early 1950s; forerunners of magnetic disk technology.
- Magnetic Disk - commercially available in mid 1950s.
- Compact Disk (CD) – introduced as a data storage medium in 1985.
- Solid-state technology – Flash drives.

Using Data for Competitive Advantage

- Data has become indispensable to every kind of modern business and government organization.
- Data, the applications that process the data, and the computers on which the applications run are fundamental to every aspect of every kind of endeavor.

Using Data for Competitive Advantage

- Data is a corporate resource, possibly the most important corporate resource.
- Data can give a company a crucial competitive advantage.
- e.g., FedEx had a significant competitive advantage when it first provided access to its package tracking data on its Web site.

Problems in Storing and Accessing Data

- Difficult to store and to provide efficient, accurate access to a company's data.
- The volume of data that companies have is massive.
- Wal-Mart estimates its data warehouse contains hundreds of terabytes (trillions of characters) of data.

Problems in Storing and Accessing Data

- Larger number of people want access to data:
 - Employees
 - Customers
 - Trading partners

- Additional issues include: data security, data privacy, and backup and recovery.

Data Security

- Involves a company protecting its data from theft, malicious destruction, deliberate attempts at making phony changes to the data.
- e.g., someone trying to increase his own bank account balance.

Data Privacy

- Ensuring that even employees who normally have access to the company's data are given access only to the specific data that they need in their work.

Backup and Recovery

- The ability to reconstruct data if it is lost or corrupted.
- e.g., following a hardware failure
- e.g., following a natural disaster

Data Accuracy

- The same data is stored several, sometimes many, times within a company's information system.
- When a new application is written, new data files are created to store its data.
- Data can be duplicated within a single file and across files.

Data as a Corporate Resource

- Data may be the most difficult corporate resource to manage.
- We have tremendous volume, billions, trillions, and more individual pieces of data, each piece of which is different from the next.

Data as a Corporate Resource

- A new kind of software is required to help manage the data.
- Progressively faster hardware is required to keep up with the increasing volume of data and data access demands.
- Data management specialists need to be developed and educated.

The Database Environment

- Database Management System (DBMS)
- New Personnel - database administrator and data management specialist
- Fast hardware
- Massive data storage facilities

The Database Environment

- Encourages data sharing
- Helps control data redundancy
- Has important improvements in data accuracy
- Permits storage of vast volumes of data with acceptable access.

The Database Environment

- Allows database queries
- Provides tools to control:
 - data security
 - data privacy
 - backup and recovery

Types of Databases and Database Applications

- Traditional Applications:
 - Numeric and Textual Databases
- More Recent Applications:
 - Multimedia Databases
 - Geographic Information Systems (GIS)
 - Biological and Genome Databases
 - Data Warehouses
 - Mobile databases
 - Real-time and Active Databases

Recent Developments (cont.)

- Social Networks started capturing a lot of information about people and about communications among people-posts, tweets, photos, videos in systems such as:
 - Facebook
 - Twitter
 - Linked-In
- All of the above constitutes data
- Search Engines- Google, Bing, Yahoo : collect their own repository of web pages for searching purposes

Recent Developments (cont.)

- New Technologies are emerging from the so-called non-database software vendors to manage vast amounts of data generated on the web:
- Big Data storage systems involving large clusters of distributed computers (Chapter 25)
- NOSQL (Not Only SQL) systems (Chapter 24)
- A large amount of data now resides on the “cloud” which means it is in huge data centers using thousands of machines.

Basic Definitions

- **Database:**
 - A collection of related data.
- **Data:**
 - Known facts that can be recorded and have an implicit meaning.
- **Mini-world:**
 - Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.
- **Database Management System (DBMS):**
 - A software package/ system to facilitate the creation and maintenance of a computerized database.
- **Database System:**
 - The DBMS software together with the data itself. Sometimes, the applications are also included.

Impact of Databases and Database Technology

- Businesses: Banking, Insurance, Retail, Transportation, Healthcare, Manufacturing
- Service Industries: Financial, Real-estate, Legal, Electronic Commerce, Small businesses
- Education : Resources for content and Delivery
- More recently: Social Networks, Environmental and Scientific Applications, Medicine and Genetics
- Personalized Applications: based on smart mobile devices

Simplified database system environment

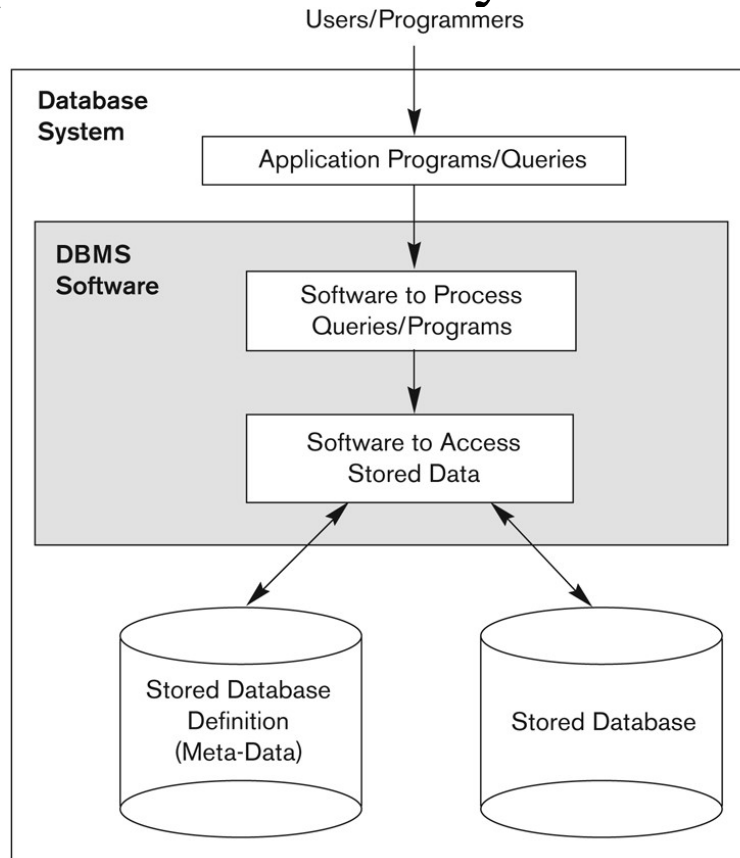


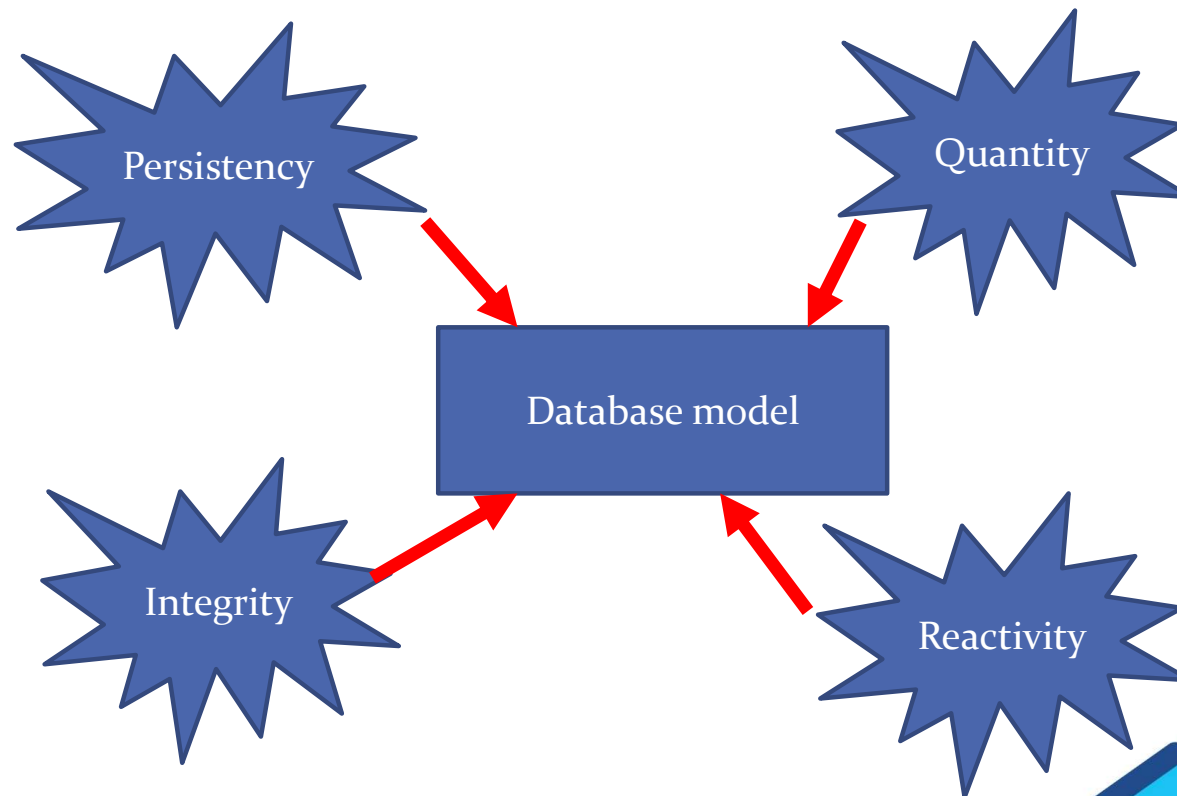
Figure 1.1
A simplified database
system environment.

Typical DBMS Functionality

- *Define* a particular database in terms of its data types, structures, and constraints
- *Construct* or Load the initial database contents on a secondary storage medium
- *Manipulating* the database:
 - Retrieval: Querying, generating reports
 - Modification: Insertions, deletions and updates to its content
 - Accessing the database through Web applications
- *Processing* and *Sharing* by a set of concurrent users and application programs – yet, keeping all data valid and consistent

Database Model Requirements

- PQRI:



Application Activities Against a Database

- Applications interact with a database by generating
 - Queries: that access different parts of data and formulate the result of a request
 - Transactions: that may read some data and “update” certain values or generate new data and store that in the database
- Applications must not allow unauthorized users to access data
- Applications must keep up with changing user requirements against the database

Additional DBMS Functionality

- DBMS may additionally provide:
 - Protection or Security measures to prevent unauthorized access
 - “Active” processing to take internal actions on data
 - Presentation and Visualization of data
 - Maintenance of the database and associated programs over the lifetime of the database application
 - Called database, software, and system maintenance

Example of a Database (with a Conceptual Data Model)

- **Mini-world for the example:**
 - Part of a UNIVERSITY environment.
- **Some mini-world *entities*:**
 - STUDENTs
 - COURSEs
 - SECTIONs (of COURSEs)
 - (academic) DEPARTMENTs
 - INSTRUCTORs

Example of a Database (with a Conceptual Data Model) (cont.)

- **Some mini-world *relationships*:**
 - SECTIONs *are of specific* COURSEs
 - STUDENTs *take* SECTIONs
 - COURSEs *have prerequisite* COURSEs
 - INSTRUCTORs *teach* SECTIONs
 - COURSEs *are offered by* DEPARTMENTs
 - STUDENTs *major in* DEPARTMENTs

- Note: The above entities and relationships are typically expressed in a conceptual data model, such as the ENTITY-RELATIONSHIP data model (see Chapters 3, 4)

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2
A database that stores
student and course
information.

Main Characteristics of the Database Approach

- **Self-describing nature of a database system:**
 - A DBMS **catalog** stores the description of a particular database (e.g. data structures, types, and constraints)
 - The description is called **meta-data***
 - This allows the DBMS software to work with different database applications.
- **Insulation between programs and data:**
 - Called **program-data independence**.
 - Allows changing data structures and storage organization without having to change the DBMS access programs.

* Some newer systems such as a few NOSQL systems need no meta-data: they store the data definition within its structure making it self describing

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerated type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits



Figure 1.3

An example of a database catalog for the database in Figure 1.2.

Main Characteristics of the Database Approach (cont.)

▪ **Data Abstraction:**

- A **data model** is used to hide storage details and present the users with a conceptual view of the database.
- Programs refer to the data model constructs rather than data storage details

▪ **Support of multiple views of the data:**

- Each user may see a different view of the database, which describes **only** the data of interest to that user.

Main Characteristics of the Database Approach (cont.)

- **Sharing of data and multi-user transaction processing:**
 - Allowing a set of **concurrent users** to retrieve from and to update the database.
 - *Concurrency control* within the DBMS guarantees that each **transaction** is correctly executed or aborted
 - *Recovery* subsystem ensures each completed transaction has its effect permanently recorded in the database
 - **OLTP** (Online Transaction Processing) is a major part of database applications. This allows hundreds of concurrent transactions to execute per second.

Database Users

- Users may be divided into
 - Those who actually use and control the database content, and those who design, develop and maintain database applications (called “Actors on the Scene”), and
 - Those who design and develop the DBMS software and related tools, and the computer systems operators (called “Workers Behind the Scene”).

Database Users – Actors on the Scene

- Actors on the scene
 - **Database administrators:**
 - Responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software and hardware resources, controlling its use and monitoring efficiency of operations.
 - **Database Designers:**
 - Responsible to define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.

Database End Users

- Actors on the scene (continued)
 - **End-users:** They use the data for queries, reports and some of them update the database content. End-users can be categorized into:
 - **Casual:** access database occasionally when needed
 - **Naïve or Parametric:** they make up a large section of the end-user population.
 - They use previously well-defined functions in the form of “canned transactions” against the database.
 - Users of Mobile Apps mostly fall in this category
 - Bank-tellers or reservation clerks are parametric users who do this activity for an entire shift of operations.
 - Social Media Users post and read information from websites

Database End Users (cont.)

- **Sophisticated:**
 - These include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities.
 - Many use tools in the form of software packages that work closely with the stored database.
- **Stand-alone:**
 - Mostly maintain personal databases using ready-to-use packaged applications.
 - An example is the user of a tax program that creates its own internal database.
 - Another example is a user that maintains a database of personal photos and videos.

Database Users – Actors on the Scene (cont.)

- **System Analysts and Application Developers**

This category currently accounts for a very large proportion of the IT work force.

- **System Analysts:** They understand the user requirements of naïve and sophisticated users and design applications including canned transactions to meet those requirements.
- **Application Programmers:** Implement the specifications developed by analysts and test and debug them before deployment.
- **Business Analysts:** There is an increasing need for such people who can analyze vast amounts of business data and real-time data (“Big Data”) for better decision making related to planning, advertising, marketing etc.

Database Users – Actors behind the Scene

- **System Designers and Implementors:** Design and implement DBMS packages in the form of modules and interfaces and test and debug them. The DBMS must interface with applications, language compilers, operating system components, etc.
- **Tool Developers:** Design and implement software systems called tools for modeling and designing databases, performance monitoring, prototyping, test data generation, user interface creation, simulation etc. that facilitate building of applications and allow using database effectively.
- **Operators and Maintenance Personnel:** They manage the actual running and maintenance of the database system hardware and software environment.

Advantages of Using the Database Approach

- Controlling redundancy in data storage and in development and maintenance efforts.
 - Sharing of data among multiple users.
- Restricting unauthorized access to data. Only the DBA staff uses privileged commands and facilities.
- Providing persistent storage for program Objects
 - E.g., Object-oriented DBMSs make program objects persistent— see Chapter 12.
- Providing Storage Structures (e.g. indexes) for efficient Query Processing – see Chapter 17.

Advantages of Using the Database Approach (cont.)

- Providing optimization of queries for efficient processing.
- Providing backup and recovery services.
- Providing multiple interfaces to different classes of users.
- Representing complex relationships among data.
- Enforcing integrity constraints on the database.
- Drawing inferences and actions from the stored data using deductive and active rules and triggers.

Additional Implications of Using the Database Approach

- Potential for enforcing standards:
 - This is very crucial for the success of database applications in large organizations.
Standards refer to data item names, display formats, screens, report structures, meta-data (description of data), Web page layouts, etc.
- Reduced application development time:
 - Incremental time to add each new application is reduced.

Additional Implications of Using the Database Approach (cont.)

- Flexibility to change data structures:
 - Database structure may evolve as new requirements are defined.
- Availability of current information:
 - Extremely important for on-line transaction systems such as shopping, airline, hotel, car reservations.
- Economies of scale:
 - Wasteful overlap of resources and personnel can be avoided by consolidating data and applications across departments.

Historical Development of Database Technology

- Early Database Applications:
 - The Hierarchical and Network Models were introduced in mid 1960s and dominated during the seventies.
 - A bulk of the worldwide database processing still occurs using these models, particularly, the hierarchical model using IBM's IMS system.
- Relational Model based Systems:
 - Relational model was originally introduced in 1970, was heavily researched and experimented within IBM Research and several universities.
 - Relational DBMS Products emerged in the early 1980s.



- Object-oriented and emerging applications:
 - Object-Oriented Database Management Systems (OODBMSs) were introduced in late 1980s and early 1990s to cater to the need of complex data processing in CAD and other applications.
 - Their use has not taken off much.
 - Many relational DBMSs have incorporated object database concepts, leading to a new category called *object-relational* DBMSs (ORDBMSs)
 - *Extended relational* systems add further capabilities (e.g. for multimedia data, text, XML, and other data types)



- Data on the Web and E-commerce Applications:
 - Web contains data in HTML (Hypertext markup language) with links among pages.
 - This has given rise to a new set of applications and E-commerce is using new standards like XML (eXtended Markup Language). (see Ch. 13).
 - Script programming languages such as PHP and JavaScript allow generation of dynamic Web pages that are partially generated from a database (see Ch. 11).
 - Also allow database updates through Web pages

Extending Database Capabilities

- New functionality is being added to DBMSs in the following areas:
 - Scientific Applications – Physics, Chemistry, Biology - Genetics
 - Earth and Atmospheric Sciences and Astronomy
 - XML (eXtensible Markup Language)
 - Image Storage and Management
 - Audio and Video Data Management
 - Data Warehousing and Data Mining – a very major area for future development using new technologies (see Chapters 28-29)
 - Spatial Data Management and Location Based Services
 - Time Series and Historical Data Management
- The above gives rise to *new research and development* in incorporating new data types, complex data structures, new operations and storage and indexing schemes in database systems.

Extending Database Capabilities (cont.)

- Background since the advent of the 21st Century:
 - First decade of the 21st century has seen tremendous growth in user generated data and automatically collected data from applications and search engines.
 - Social Media platforms such as Facebook and Twitter are generating millions of transactions a day and businesses are interested to tap into this data to “understand” the users
 - Cloud Storage and Backup is making unlimited amount of storage available to users and applications

Extending Database Capabilities (cont.)

- Emergence of Big Data Technologies and NOSQL databases
 - New data storage, management and analysis technology was necessary to deal with the onslaught of data in petabytes a day (10^{15} bytes or 1000 terabytes) in some applications – this started being commonly called as “Big Data”.
 - Hadoop (which originated from Yahoo) and Mapreduce Programming approach to distributed data processing (which originated from Google) as well as the Google file system have given rise to Big Data technologies (Chapter 25). Further enhancements are taking place in the form of Spark based technology.
 - NOSQL (Not Only SQL- where SQL is the de facto standard language for relational DBMSs) systems have been designed for rapid search and retrieval from documents, processing of huge graphs occurring on social networks, and other forms of unstructured data with flexible models of transaction processing (Chapter 24).

When not to use a DBMS

- Main inhibitors (costs) of using a DBMS:
 - High initial investment and possible need for additional hardware.
 - Overhead for providing generality, security, concurrency control, recovery, and integrity functions.
- When a DBMS may be unnecessary:
 - If the database and applications are simple, well defined, and not expected to change.
 - If access to data by multiple users is not required.
- When a DBMS may be infeasible:
 - In embedded systems where a general purpose DBMS may not fit in available storage

When not to use a DBMS

- When no DBMS may suffice:
 - If there are stringent real-time requirements that may not be met because of DBMS overhead (e.g., telephone switching systems)
 - If the database system is not able to handle the complexity of data because of modeling limitations (e.g., in complex genome and protein databases)
 - If the database users need special operations not supported by the DBMS (e.g., GIS and location based services).

References

- Ramez Elmasri, Shamkant Navathe; “Fundamentals of Database Systems”, 6th Ed., Pearson, 2014.
- Mark L. Gillenson; “Fundamentals of Database Management Systems”, 2nd Ed., John Wiley, 2012.
- Universität Hamburg, Fachbereich Informatik, Einführung in Datenbanksysteme, Lecture Notes, 1999